Dissertation: Quality-Adaptive Multi-Modal Face Recognition with Streaming Prototype Learning and Cross-Modal Evaluation

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Chapter 1 — Introduction

- Real-world face recognition fails due to quality variation, modality mismatch, and static galleries
- Gap: no existing system jointly addresses all three
- Research questions:
  1. How can quality variation across modalities be handled adaptively?
  2. How can identity galleries update incrementally without full retraining?
  3. How should cross-modal performance be standardized and evaluated?
- Contributions (4 clear bullets)
- Dissertation outline

Chapter 2 — Background and Related Work

- Face recognition pipeline overview
- Loss function evolution: CosFace → ArcFace → AdaFace
- Quality-aware recognition: FaceQnet, SER-FIQ, AdaFace quality proxy
- Multi-modal fusion: RGB-D, NIR-VIS, thermal approaches
- Continual/incremental learning: catastrophic forgetting, prototype networks, replay buffers
- Cross-spectral evaluation: existing protocols and their fragmentation
- Gaps table: what exists vs. what this dissertation addresses

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Chapter 3 — Quality Assessment Module

Design

- Per-image quality estimator for each modality independently
- RGB quality: pose deviation, blur, occlusion, resolution

- NIR/Depth quality: sensor noise, alignment artifacts, saturation
- Architecture: lightweight CNN head on top of frozen backbone, outputs scalar score ∈ [0,1]
- Use AdaFace's quality proxy as baseline — extend to non-RGB modalities

Experiment 1 — Quality Score Validation

- **Dataset:** LFW with synthetically degraded variants (blur, noise, occlusion, low-res)
- **Protocol:** correlate predicted quality score with actual verification accuracy drop
- **Metrics:** Spearman correlation between quality score and TAR@FAR
- **Baseline:** SER-FIQ, FaceQnet
- **Expected result:** multi-modal quality estimator correlates more strongly with cross-modal accuracy than single-modal baselines

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Chapter 4 — Quality-Adaptive Multi-Modal Fusion

Design

- Dual-stream encoder: one per modality (ResNet-50 backbone each)
- Quality scores from Chapter 3 gate the fusion weights dynamically
- Fusion mechanism: attention layer where attention weights = f(quality scores)
- Trained end-to-end with ArcFace loss on fused embedding
- Key property: degrades gracefully — if one modality is missing/low quality, system relies on the other

Experiment 2 — Fusion Ablation

- **Dataset:** IIIT-D RGB-D, CurtinFaces
- **Protocol:** standard identification
- **Conditions:**
  - RGB only
  - Depth only
  - Fixed equal fusion
  - Quality-adaptive fusion (proposed)
- **Metrics:** Rank-1 accuracy per condition
- **Expected result:** quality-adaptive fusion outperforms fixed fusion, especially under degraded conditions

Experiment 3 — Robustness Under Degradation

- **Dataset:** CASIA NIR-VIS 2.0 with artificially degraded NIR inputs
- **Protocol:** 10-fold identification
- **Conditions:** degrade NIR quality progressively (5 levels), measure Rank-1 at each level
- **Metrics:** Rank-1 vs. degradation level curve
- **Baseline:** fixed fusion, RGB-only
- **Expected result:** system degrades more gracefully than fixed fusion as NIR quality drops

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Chapter 5 — Streaming Prototype Learning

Design

- Each identity represented by a **prototype** — momentum-updated mean embedding in feature space
- New samples update the prototype via exponential moving average: `p_i ← α·p_i + (1−α)·e_new`
- New identities: initialize prototype from first sample, add to memory bank
- Forgetting prevention: reservoir sampling replay buffer — periodically replay stored embeddings
- No full retraining required — gallery updates at inference time

Experiment 4 — Streaming Identity Addition

- **Dataset:** IJB-C split into 5 sequential temporal batches
- **Protocol:** add identities batch-by-batch, evaluate Rank-1 after each batch
- **Conditions:**
  - Static gallery (no update)
  - Full retraining after each batch
  - Streaming prototype learning (proposed)
- **Metrics:** Rank-1 accuracy over time, wall-clock update time
- **Expected result:** proposed approach matches full retraining accuracy at a fraction of compute cost

Experiment 5 — Catastrophic Forgetting Measurement

- **Dataset:** IJB-C sequential split
- **Protocol:** after adding new identities, re-evaluate accuracy on original identities
- **Metrics:** Backward Transfer (BWT) — measures how much old identity accuracy drops
- **Baseline:** naive prototype update without replay buffer

- **Expected result:** replay buffer significantly reduces BWT, approaching full retraining stability

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Chapter 6 — Cross-Modal Evaluation Protocol

Design

Protocol Definition — 4 standardized cross-modal scenarios:

Scenario	Train	Test
1	RGB	NIR
2	NIR	RGB
3	RGB + NIR	RGB only
4	RGB + NIR	NIR only

- Fixed FAR thresholds: 1e-3, 1e-4 (consistent across all scenarios)
- Required reported metrics: Rank-1, TAR@FAR, DIR@FAR=1%
- Demographic subgroup reporting required
- Directly addresses evaluation fragmentation identified in related work — proposes the standard the field lacks

Experiment 6 — Protocol Benchmarking

- **Dataset:** CASIA NIR-VIS 2.0, TUFTS Thermal-VIS
- **Baselines evaluated under protocol:** ArcFace (RGB-only), coupled feature learning, domain adaptation, full proposed system
- **Metrics:** all 4 protocol scenarios × 3 metrics
- **Expected result:** proposed system shows smallest performance drop across modality mismatch scenarios

Experiment 7 — Fragmentation Analysis

- Take 10 published methods, attempt to evaluate all under unified protocol
- Show how many can be directly compared vs. how many require assumptions
- **Output:** quantitative evidence that the proposed protocol resolves the comparability problem

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Chapter 7 — Full System Evaluation

Experiment 8 — End-to-End System Benchmark

- **Datasets:** IJB-C, CASIA NIR-VIS 2.0, CurtinFaces, RFW
- **Protocol:** cross-modal protocol from Chapter 6
- **Conditions:** full system vs. each ablated component
- **Metrics:** Rank-1, TAR@FAR@1e-4, DIR@FAR=1%

Experiment 9 — Fairness Audit

- **Dataset:** RFW (4 ethnic subgroups)
- **Metrics:** TAR per subgroup, FMR disparity, FNMR disparity
- **Baseline:** ArcFace baseline
- **Expected result:** quality-adaptive weighting reduces demographic performance gap by reducing reliance on low-quality inputs that disproportionately affect certain demographics

Experiment 10 — Deployment Readiness Scoring

- Formalize deployment readiness framework from related work
- Score proposed system and 5 SOTA baselines across all dimensions:

Dimension	Weight
Accuracy	25%
Fairness	20%
Adversarial robustness	20%
Open-set capability	20%
Deployment feasibility	15%

- **Output:** ranked deployment readiness table + radar charts per method

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Chapter 8 — Results Summary

Table	Content	Experiment
Table 1	Quality score correlation vs. baselines	Exp. 1
Table 2	Fusion ablation — Rank-1 per condition	Exp. 2
Table 3	Degradation robustness curves	Exp. 3
Table 4	Streaming accuracy over time vs. baselines	Exp. 4

Table	Content	Experiment
Table 5	Backward Transfer scores	Exp. 5
Table 6	Cross-modal protocol benchmark — all 4 scenarios	Exp. 6
Table 7	Full system vs. SOTA on IJB-C and CASIA NIR-VIS 2.0	Exp. 8
Table 8	Fairness audit — TAR per demographic subgroup	Exp. 9
Table 9	Deployment readiness scores	Exp. 10

Chapter 9 — Discussion

9.1 Quality Adaptation

- Does quality-adaptive fusion actually help, or does the backbone learn to ignore low-quality inputs anyway?
- At what quality threshold does switching modalities become beneficial?
- Failure cases: both modalities simultaneously degraded

9.2 Streaming vs. Static Tradeoff

- How many streaming updates before prototype drift becomes a problem?
- Is the replay buffer size a critical hyperparameter?
- Comparison to continual learning literature — where does the approach sit?

9.3 Cross-Modal Protocol

- Which modality mismatch scenario is hardest? Why?
- Does the proposed protocol expose weaknesses in SOTA that existing protocols hide?
- Limitations: protocol only covers NIR-VIS and RGB-D — thermal and synthetic modalities need future work

9.4 Fairness Implications

- Does quality-adaptive weighting inadvertently introduce new demographic biases?
- If NIR sensors perform differently across skin tones, does trusting NIR quality scores propagate that bias?

9.5 Limitations

- Evaluated on existing public datasets — real deployment conditions may differ

- Streaming protocol assumes identity labels are available at update time
- Quality estimator trained on synthetic degradations — may not generalize to all real-world artifacts
- Cross-modal protocol covers 4 scenarios — not exhaustive

Chapter 10 — Conclusion

- Summary of each component's contribution
- Answer to each research question
- Broader impact: framework applicable beyond face recognition to any multi-modal biometric system
- Future work:
  - Unsupervised streaming (no identity labels at update time)
  - Extending protocol to video modality
  - Privacy-preserving prototype storage

Experiments at a Glance

#	Experiment	Chapter	Datasets	Key Metric
1	Quality score validation	3	LFW degraded	Spearman correlation
2	Fusion ablation	4	IIIT-D RGB-D, CurtinFaces	Rank-1
3	Degradation robustness	4	CASIA NIR-VIS 2.0	Rank-1 vs. degradation level
4	Streaming identity addition	5	IJB-C sequential	Rank-1 over time
5	Forgetting measurement	5	IJB-C sequential	Backward Transfer
6	Cross-modal protocol benchmark	6	CASIA NIR-VIS 2.0, TUFTS	TAR@FAR all 4 scenarios
7	Fragmentation analysis	6	10 published methods	Comparability score
8	End-to-end benchmark	7	IJB-C, CASIA, CurtinFaces	Rank-1, TAR@FAR, DIR
9	Fairness audit	7	RFW	TAR per subgroup

#	Experiment	Chapter	Datasets	Key Metric
10	Deployment readiness	7	All above	Composite score